



Tobias Friedrich

AI Architect & Engineering Lead

Summary

Hands-on architect and tech lead with 7+ years of experience shipping scalable, AI-driven products. I specialize in building end-to-end solutions, from cloud infrastructure to agentic AI systems, that solve real-world business problems. Passionate about leading small, agile teams to deliver high-impact results in fast-paced environments.

Featured Projects

- Shipped Q1 2026** **Genie Platform: Autonomous Agent-driven Development, Internal Pilot**
- Architected and led the development of a platform using supervised autonomous agents to automate the software development lifecycle, from requirements engineering to testing.
 - Engineered a system leveraging RAG and LLMs to analyze the knowledge base and drive agentic workflows, resulting in a 30
 - *Tech Stack: Agentic System, LangGraph, Langfuse, RAG, LLMs, Python, React, Azure.*
- Prototype Q4 2025** **AI-Powered Automated Test Generation, Internal Tool**
- Designed a system that slashed test creation time by 80% by using RAG to parse requirements and automatically synthesize test cases.
 - *Tech Stack: Python, RAG, LLMs, Vector Databases, Azure, React.*
- Shipped Q2 2024** **Global Skills Platform, Global Rollout**
- Led the technical rollout of a cloud-native skills platform to over 200,000 users, achieving 99.9% uptime and seamless integration with multiple enterprise SaaS products.
 - Enabled personalized career development, leading to a 15
 - *Tech Stack: SaaS, REST APIs, Event-Driven Architecture, Python, Go, Kubernetes.*

Experience

- May 2023 – Present** **Tech Lead, AI & Cloud Solutions, Robert Bosch GmbH, Stuttgart, Germany**
- Led a team of 3 engineers to ship a global, AI-powered skills platform for 200,000+ users, resulting in a 20
 - Owned the architecture of scalable cloud solutions on Azure, reducing cloud costs by 15
 - Organized and led two enterprise-wide AI hackathons, fostering innovation and generating over 10 promising new product concepts for the company.
 - Acted as the go-to AI consultant for enterprise clients, translating business needs into technical requirements and product roadmaps that secured three major client contracts.

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Oct 2017 – **Software Engineer, Autonomous Driving**, *Bosch Engineering GmbH*, Stuttgart, Germany
Apr 2023

- Deployed production-grade computer vision algorithms for autonomous driving; improved detection accuracy to 99.2
- Built a Python-based test automation framework from the ground up; reduced testing cycles by 40
- Designed and managed CI/CD pipelines for robust embedded software deployment, decreasing deployment failures by 25

Technical Proficiencies

AI/ML Engineering Generative AI (LLMs, RAG, Diffusion Models), Agentic Systems (LangGraph, Langfuse), Deep Learning (PyTorch, TensorFlow), NLP, Computer Vision, Knowledge Graphs

Cloud & Architecture Solution Architecture, Microservices, Event-Driven Systems, API Management (REST), Azure, Docker, Kubernetes, Openshift, High-Availability Systems

Languages & DevOps Python (Expert), Go (Advanced), C/C++ (Advanced), React, SQL, Bash, CI/CD (Jenkins), Git, Jira

Education

2020 – 2023 **M.Sc. Engineering Informatics**, *University of Augsburg*, Germany
Focus: Deep Learning & Artificial Intelligence

2014 – 2017 **B.Eng. Mechatronics**, *DHBW Stuttgart*, Germany
Partner Company: Robert Bosch GmbH